AaS! 282: What Happens When You Shine a Light on Nothing?

Podcast: Ask a spaceman

AI Executive Summary

Core Thesis

Empty space is fundamentally "full," composed of quantum fields and virtual particles that fluctuate constantly due to Heisenbergâ€™s Uncertainty Principle. While typically invisible, this vacuum is reactive; light interacts with it through scattering, polarization effects mediated by magnetic fields (vacuum birefringence), and potential particle creation under extreme electric field intensities (Schwinger effect). Understanding these interactions shifts the paradigm from a passive void to an active medium that can be manipulated by extreme astrophysical or laboratory conditions.

Key Arguments & Data

- **Nature of Vacuum:** The vacuum is not "nothingness" but the lowest energy state (ground state) of permeating quantum fields. Fields possess intrinsic jitter; they cannot be perfectly still, guaranteeing non-zero vacuum energy.
- **Virtual Particles:** These are bookkeeping devices representing transient excitations in quantum fields that appear and disappear within limits set by conservation of energy, rather than literal matter populating the void.
- **Light-Light Interaction (Photon-Photon Scattering):** Photons interact via virtual particles (e.g., positron-electron pairs). In a vacuum, two photon beams rarely scatter and emit new photons in different directions (observed at the Large Hadron Collider in 2017), with a cross-section billions of times rarer than standard particle interactions.
- **Vacuum Birefringence:** Strong magnetic fields align virtual particles into structured orientations. This turns the vacuum into an optical medium, where light polarized parallel to the field travels differently from light perpendicular to it.

    - **Magnetic Threshold:** Fields must exceed ~10 trillion times Earth's magnetic strength (found in Magnetars) to significantly affect vacuum polarization; evidence comes from X-ray radiation observations near magnetars.
- **Schwinger Critical Field:** A strong enough electric field can "boil" the vacuum by pulling virtual pairs apart before they recombine, promoting them into real particles. This requires concentrated energy density not yet achieved fully by current laser technology.
    - **Current Limitations:** State-of-the-art facilities (e.g., Extreme Light Infrastructure, X-watt Center) are ~1000 times weaker than the Schwinger limit.
    - **Future Potential:** Future ultra-powerful lasers could reach the critical threshold in nanoseconds or microseconds to directly extract particles from the vacuum.

Actionable Takeaways

- **Monitor High-Energy Laser Development:** Keep track of advancements at facilities like the Extreme Light Infrastructure (ELI), as they are the primary means for approaching the Schwinger limit and potentially inducing vacuum particle creation.
- **Update on Magnetar Physics:** Follow astrophysical observations regarding Magnetars to see if experimental data on X-ray polarization further confirms the theoretical predictions of vacuum birefringence in space.
- **Stay Informed on Quantum Electrodynamics (QED):** The episode highlights the distinction between standard model expectations and extreme field physics; follow research into "light-by-light scattering" as it remains a leading candidate for direct vacuum interaction beyond the current observational limits.
- **Engage with Scientific Inquiry:** Use the channel provided in the transcript (AskA Spaceman emails/website) to query similar thought experiments about high-energy physics, as the show encourages audience interaction on complex scientific topics.

Full Transcript

Take out a flashlight. Shine it on something. A wall, your face, that breeze sitting on the counter, the mirror. The destination is not the point of this episode. It's the journey. What do you see? Well, you see the light reflecting off whatever it is you're pointing at and you also see the beam itself because of all the dust scattered in the air that's providing millions of little micro reflections back at you. So let's put that light in the vacuum of space and repeat the experiment.

This time you don't see the beam of light because there's nothing to bounce the light off of or scatter or interact with on the light's way to the thing you're pointing at. Or is there? I don't know why I'm asking rhetorical questions because obviously there's more to this scenario, otherwise I wouldn't have devoted an entire episode to the topic. And the reason I get to devote an entire episode to this topic is because while we're all taught in kindergarten that the vacuum of space is perfectly empty, we're a bit past kindergarten here. Maybe Maybe second grade. And I'm not talking here about wandering bits of dust or stray cosmic rays or hydrogen atoms or even dark matter. I'm talking about the very nature of the vacuum itself. Empty space is anything but. And we've gone through this before so I'll make it brief, but we need a refresher here so we understand just what the heck can go wrong if we shine a light onto the vacuum. The old picture is that empty space is the absence of stuff and that's it. Pure, unadulterated.

nothingness. The very kind of vacuum that Aristotle abhorred and therefore concluded that nature did as well. But this isn't Aristotle show. It's not even Newton's show. This is Heisenberg's show. It's Feynman's show. Well, actually, it's my show. But this episode is based more on what they taught us about the vacuum. And they taught us that every single particle is really a vibration or the technical term excitation of an underlying field. There's a electron field that permeates all of space of time and when little bits of it vibrate, we get electrons. There's a photon field, we call it the electromagnetic field, and when little bits of it vibrates, we get photons. There's a top quark field, there's a neutrino field, and presumably there are

fields for the dark matter, whatever the heck that is. And the study of these fields is conveniently enough called quantum field theory.

And the fields exist everywhere. They permeate space and time. My favorite analogy is oil and vinegar soaked in bread. They're everywhere, even where the particles aren't. Because there's always a field, but only sometimes a bit of the field gets enough energy to excite where we point at and say, hey, there's a particle. But when the particle goes away, it just means the vibrations in the field have slipped below the threshold needed to call itself a particle. I know this sounds all hippie like here we are with vibrations. Maybe I'm going to talk about the crystals. No, no, no, it's it's real physics. It's quantum field theory. It is the theory of quantized fields that give rise to particles and, you know, all material existence. Anyway, what we need to care about the fields is that they're in the vacuum. Even when there are no particles, there is still a field. So the vacuum is not empty. It is full of quantum fields. And what we call the vacuum is really the ground state of all those fields is their lowest possible energy configuration. But this is quantum mechanics. Heisenberg's uncertainty principle forbids things from being perfectly still. Even in this ground state, even in this lowest possible energy, you get rid of all the particles, you calm everything down, you like a librarian on a little patch.

A vacuum, you get the quantum fields to go settle down, but like a toddler, they just have a little bit of vibration. Even when they're sitting in a chair, they're just just a little bit of wiggle that you can't get rid of no matter how much you shush them. Quantum fields, their lowest possible energy is not zero energy. Even in the ground state, fields fluctuate. Vacuum, the vacuum of space time. has a sort of intrinsic jitter to this. Now, we need to talk about this intrinsic jitter, we need to talk about this fundamental vibration that that permeates the entire cosmos, even in the vacuum of space, because that's what's going to interact with our light when we shine light in a vacuum. Now, normally, when I talk about quantum fields, I despise introducing the concept of virtual particles.

Virtual particles are this idea that at the smallest levels of existence that there are particles popping in and out of existence that they appear and then they disappear and that they can exist just briefly long enough before

anyone notices before any there before there's any violation of conservation of energy and then they return back into the vacuum and they do this by spawning in pairs of particle antiparticle pairs so that you get, like, a top quirk and a top anti-quirk, or a positron and an electron, they spawn together out of nowhere, unabidden, unasked for, and they borrow a little bit of energy from the universe to do so, and then they recolide, vanish, and the energy goes back. Honestly, this picture is a poor description of what's really going on, and introduces more problems and solutions, like, for example, what the heck is virtual about virtual particles.

And the truth is that virtual particles is a bookkeeping device, not a literal cloud of stuff that exists at subatomic scales. But it's a picture that is occasionally useful. And as much as I hate it, it's going to be useful for us today. So virtual particles, you finally get your moment on the show after this commercial break. Now that we have the right setup, we can have some fun. Because we get to reframe what light shining through the vacuum really is. Photons themselves are vibrations slash excitation slash wiggles of the electromagnetic field. That electromagnetic field is moving through all sorts of other fields on its way to its destination, so we get to ask, does one kind of quantum field get to talk to other kinds of quantum fields? And the answer is yes, otherwise it wouldn't exactly be a useful theory of physics.

And we can start our first game just asking about the photons themselves. Can two beams of light talk to each other in the vacuum with nothing else going on? In the universe of Newton and even Maxwell, the answer is always no, light beams just add together where they meet and then subtract when they go away. Done and done. But in the quantum field theory universe, photons are always interacting with the virtual particles of the vacuum. They're constantly changing places, swapping in and out, transforming into each other and back again. You imagine this photon, which is itself and wiggle and excitation of the electromagnetic field as it's propagating through the vacuum, you can imagine this sea of virtual particles and you can imagine the photons themselves as virtual particles. And it's not just a stream of photons going down a pipe. They like hit a little virtual particle, they ricochet off of it, they bounce off of it.

They interact with it and then the virtual particle goes away and the light goes on. But there's all sorts of shuffling happening at subatomic scales. And when two photons meet, remember, they themselves are excitations. They themselves are bundles of virtual particles. Most of the time nothing happens. But sometimes really, really, really rarely, I can't even begin to describe just how rare it is. The photons interact. with a pair of virtual particles. And then those pairs of virtual particles create new photons that go off in new directions. The idea is if you have two beams of light interacting, then really, really, really occasionally, rarely, one beam of light over here interacts with a positron and electron, say a pair of virtual particles. And then the other beam of light interacts with its own pair of virtual particles, then all those four virtual particles get mixed up together, and then they re-collapse, they re-collide, they emit light, but the light goes off in a brand new direction. This was predicted all the way back in 1935, but the chances of it happening, what physicists call the cross-section, are like a billion times less often than typical particle interactions, and yeah.

In a typical flashlight beam, there are, I don't even know how many photons, so technically this is happening every time you go looking for something in the dark, the amount of times is so slow. And the effect is rather unremarkable because if one out of a billion photons goes straight, you don't really notice, then it took all the way up until 2017 in the Large Hadron Collider before we were able to finally detect it. But that's two photons interacting in the vacuum of space. What if we just have one? Two photons proves that there is something funny happening with the vacuum of space that light does touch and we can extract some interesting mechanics out of it. What if we just have one beam of light? Well, the downside of this question is unlike the two light beam scenario, we don't have an experiment that could just pull this off. But the upside is that nature does it for us. Now this particular trick has a name.

It's a bit of a mouthful, but it's a fun mouthful. It's called Patreon. Patreon.com slash PM Sutter is how you can keep the show going and I truly appreciate it. That's patreon.com slash PMSU TT ER. It's my name, by the way, if you haven't caught that yet. The actual term is called by refringence. It's, well, okay, we need to talk about polarization. You know, polarization of light. You look at light. Light is waves of electricity and

magnetism. That's one way to view it. That's the classical view. I know we've already talked about light being photons and virtual photons working at we're going to shift pictures here a little bit in this episode so we can get to the juicy bits. You can imagine light as waves of electricity and magnetism. This is the classical Maxwell kind of stuff.

And these waves are perpendicular to each other so the electric waves are wiggling one way and then the magnetic waves are wiggling the other and then they all wiggle together to move forward. And that orientation of the electric field waves, the orientation of these waves is called the polarization. This is how polarized sunglasses are so amazing because light reflecting off a shallow angle off of a shiny reflective surface. tends to have polarization going one way. And so you have glasses that specifically block that kind of polarization out. And I'll let you see beyond the glare. And some materials some stuff affects one polarization more than the other like polarizing glasses. That's a pretty big effect. Polarizing glasses only allow one kind of polarization through if you're the other kind of polarization, you're out. It turns out that your polarized glasses are birefringent. They, uh, fringe-rate. It's not word, but okay, work with me. Different polarizations of light. And I guess birefringent glasses doesn't exactly sell well on marketing copies. So polarized glasses, it is. Anyway, what if I told you that I can make the vacuum of space act like you're polarized glasses? All it takes is a magnetic field. Not just any magnetic field, but a real whopper of one. And What do supergiant magnetic fields do while they make electric systems line up with them? Like electric systems like you guessed it, the virtual particles popping in and out of the vacuum of space. If you have a random patch of space, the virtual particles that pop into existence, they're always pairs, opposite charges, and they're just all over the place, all sorts of different orientations, directions, angles, it's just a mess.

But if you put a strong magnetic field there, then when they pop into existence, then the positive charges are going to go one way and the negative charges are going to go another. And so they line up a strong enough magnetic field lines up the virtual particles. That means that when light passes through this vacuum filled with strong magnetic fields, the bits of light that happen to be parallel to the magnetic field travel at one speed. They have the virtual particles lined up with them, helping them along,

while the opposite happens to the perpendicular field. So a super strong magnetic field turns the vacuum of space and all these virtual particles into polarized glasses. What kind of magnetic field do we need? Well, something like 10 trillion times stronger than the Earth's magnetic field will do.

Now, where the heck are we going to get a magnetic field that strong? Oh yes, my friends, it's the magnetars. This happens in nature with a rapidly spinning neutron star. They can get magnetic fields tens of trillions of times stronger than the Earth, the strongest magnetic fields in the universe. And you betcha those magnetic fields around magnetars, polarize the vacuum of space, they make the virtual particles all line up. And then when light passes through it, It's just like light passing through polarized glasses. Now we haven't exactly observed this, but there's a growing body of evidence gathered by watching X-ray radiation come out of the magnetars and comparing the polarization to the magnetic field structure that this is a real thing. So light can touch light in the vacuum and a strong magnetic field can bend the vacuum in a way that light can feel. But what about the ultimate question? Can we get light to interact with the vacuum itself? What would it take?

The answer is obvious, my friends. More power. Magnetic fields can grab onto those little virtual particles and in their brief existence force them to align with the field. They can polarize them. That's what magnetic fields do. They twist things around. But what about electric fields? Well, those push things around. And if I have a pair of virtual particles of opposite charge, imagine for just a tiny moment freezing time before they disappear. and applying an electric field to that. Well, if I were to do that, then the positive charge would go one way and the negative charge would go the other. But then they would blink out of existence and all the fun would be over. Wait, hold on a second. What if instead of a regular electric field, we had a really, really big electric field? For the virtual particles to disappear, they would need to meet. And if my field was strong enough, well, they couldn't meet, right?

This is called the Schwinger critical field named after Julian Schwinger, one of the founders of quantum field theory. And it's called critical because

well, just like when a nuclear power plant goes critical. Well, interesting things happen. The idea here is that you can energize the raw vacuum of space. If you apply a strong enough electric field, then those virtual particles that appear get pulled apart. faster than they can recombine. There's no violation of energy conservation here. You have to put energy into the vacuum. And when you do, you get to promote these virtual particles into real particles. You're literally pumping energy into the vacuum. So it's all good. Yes, there is a violation of particle number. You can have a vacuum of space with zero particles, apply an electric field, and then all these particles start whizzing out of that.

But that's par for the course of quantum field theory. It turns out number of particles is not a conserved quantity. We can create and destroy particles at will in this universe. What matters is energy. It's like boiling the vacuum here. And once you do, and to get this to work, you need extremely strong electric fields all concentrated into a super duper tiny volume like No magnet are is going to be able to do this. Yes, it can play games with magnetic fields and it can get light to twist around inside the vacuum. But to get this effect to reach a schwinger critical field with strong enough electric fields, you need super duper strong electric fields all concentrated in a tiny, tiny, tiny volume. And our best way to get super strong electric fields compressed into a tiny, tiny volume is to use an ultra powerful laser.

And we got a bunch of them. Man, we got powerful lasers all over the place. We've got the extreme light infrastructure, the x-watt center for extreme light studies, the station of extreme light. Basically, if you see name, light and extreme in the same label, then you've got a vacuum boiler operating. These facilities power up stupendously strong lasers, usually many, many of them. have them all shine their light into a single spot to see what kind of crazy energies we can get. As you might imagine, these are rather hungry facilities electricity wise. And so the these lasers fire for like a nanosecond, maybe a microsecond if you're lucky, we can't sustain that kind of power output for long. And I have to admit, even our modern facilities at the time of recording of this episode are about 1000 times weaker.

than the Schwinger critical field. They're still too weak to boil the vacuum. But we largely suspect it's a matter of when and not if. So if you're listening to this episode, a generation from now, maybe we've already done this. Maybe we accidentally blew up the universe. I don't know. I don't. That's not what's going to happen. What's going to happen is if you pour enough light into the vacuum of space, that light will drive apart the virtual particles faster than they can. recombine. And so you will shine a light on when you like shine a laser on wood, smoke starts coming out. If you shine a powerful enough laser on vacuum, the particles start coming out. And to me, that counts as light interacting with the vacuum. And we've got three is like an interact interact with itself in the vacuum of space mediated by the virtual particles that are constantly popping in and out of existence.

Light can get bent and polarized in the presence of strong magnetic fields using nothing but the vacuum itself because strong magnetic fields polarize the vacuum of space and then light responds to that polarization. And then if you have strong enough light itself, you get to promote virtual particles into real particles. So there you go. If you want light to interact with the vacuum, if you shine light on nothing itself, Well, nothing happens most of the time, but if you have a whole lot of light, then some really wild stuff happens, but I still don't like virtual particles, but that's the subject for another episode. Thank you for the questions from P.T. and at Dragon Maurizio for the questions that led to today's episode. Please keep those comments and reviews coming on your favorite podcasting platform. It really helps the show visibility and of course.

Send me more questions. Ask a spaceman at gmail.com or the website. Ask a spaceman.com and patreon.com slash PM Sutter is how you can keep this show going. I'd like to thank my top contributors this month. They are Justin G, Chris L, Alberto M, Corey D, Michael P, Naila Samar, Joshua Scott M, Rob H, Scott M, Lewis M, John W, Alexis Gilbert M, Rob W, Jessica M, Jules R, JML, David S, Scott R, Heather, Mike S, Pete H, Steve S, Lisa R, Kevin B, Eileen G, Dev A, Michael J, Phil Bel and Stephen B. Patreon.com slash PM Sutter and I will see you next time for more complete knowledge of time and space.